Findings and continuation general physics and model training setup

Sl	Author	link	Heading	Findings/possible uses
1	**Ansys Fluent Theory Guide (2011)**			to validate the theoretical foundation of the simulations your guide ran. It details the enthalpy-porosity formulation used to resolve the solid-liquid phase change (mushy zone) where porosity shifts from 0 to 1, as well as the Boussinesq approximation for natural convection
2	**Kamkari, B., & Amlashi, H.J. (2017)**	**https://doi.org/10.1016/j.icheatmasstransfer.2017.07.023**	**Numerical simulation and experimental verification of constrained melting of phase change material in inclined rectangular enclosures**	Highlights the fluid dynamics, natural convection, and geometric sensitivity (like inclination and shape constraints) of PCMs melting within rectangular enclosures. This supports your exploration of how varying aspect ratios (AR = 0.3, 0.4, 0.5) impact thermal momentum and heat transfer within the container
3	**Ben Hamida, M. B., et al.**	**https://doi.org/10.1016/j.icheatmasstransfer.2025.109781**	**Intelligent design framework for finned solar air heaters: A synergy between PSO/GA-tuned MLPNN**	Demonstrates that applying machine learning (multilayer perceptrons) to thermo-hydraulic data provides accurate predictive surrogate models for systems involving complex conjugate heat transfer. You can cite this to back up the robustness of data-driven methods in predicting thermal behavior.
4	**Basu, A., et al.**	**https://doi.org/10.1063/5.0287985**	**Modeling microchannel flow of Phan–Thien–Tanner fluids using hybrid skip-physics informed neural network.**	Highlights the capability of Physics-Informed Neural Networks (PINNs) in solving complex fluid dynamics and heat transfer equations mesh-free. It demonstrates that embedding governing physical laws into the loss function greatly improves predictive accuracy over purely data-driven networks, especially when handling non-linear interactions and sparse data. This perfectly backs your choice to use a Huber-MSE composite loss function constrained by

				thermodynamic boundaries.
5	Shen et. al	https://doi.org/10.1016/j.aitf.2024.100001	**Machine learning for predicting the PCM melting process in a rectangular enclosure energy storage.**	This paper explicitly proves that Artificial Neural Networks (ANN) can successfully map complex PCM phase-change dynamics (like melting front evolution and temperature distribution). It found that ML models reduce computational simulation times from several days down to mere minutes while retaining high accuracy (R^2 values ~ 0.99), perfectly supporting your argument that your Bi-LSTM/PINN model resolves the computational latency of traditional CFD solvers
6	Nandi and Biswas	https://doi.org/10.1016/j.est.2024.115076	**Melting dynamics and energy efficiency of nano-enhanced phase change material (NePCM) with graphene, Al2O3, and CuO for superior thermal energy storage (TES).**	Groundwork and Basis of the Paper
7	**Abdelkareem, M.A., et al.**	**https://doi.org/10.1016/j.est.2022.104385**	**Battery thermal management systems based on nanofluids for electric vehicles.**	Provides a comprehensive review of the safety risks (thermal runaway) caused by high heat generation rates in electric vehicle batteries. It highlights that integrating PCMs with thermal enhancers (like nanoparticles/nanofluids) is highly effective in controlling maximum temperature rise and improving temperature uniformity during discharge cycles
8	**Sultan, H.S., et al.**	**https://doi.org/10.1093/jcde/qwae020**	**Revolutionizing latent heat storage... in vertical shell-and-tube**	This paper examines the impact of various parameters, including frames, zigzag number, and enclosure shape, on the solidification

			systems.	process and thermal energy storage rate of a vertical phase change material (PCM) container. The study also assesses the effects of the flow rate of the heat transfer fluid as well as changing the materials of the PCM between RT35 and RT35HC. In addition, the study compares the framed versus unframed systems and, subsequently, the best case was tested with various zigzag pitch numbers before changing the zigzag-shaped structure to arc and reversed-arc. The findings are examined by contrasting the different scenarios' liquid fractions, temperature distributions, solidification rates, and heat storage rates. The results show that the framed geometry is 66% faster to reach the target temperature compared with the unframed geometry and employing a zigzag enclosure in a PCM can significantly improve the solidification time and heat recovery rate. As the number of pitches in the zigzag enclosure increases, the improvement rate decreases but still improves the solidification time and heat recovery rate. The reversed-arc-shaped structure has the best performance compared with the other undulated surfaces. For the system with RT35HC, the discharge time is 55% higher compared with that of the system with RT35, while the discharge rate is 8.2% higher for the former during the first 3000 s of the discharging process.
9	**Khedher, N.B., et al.**	**https://doi.org/10.1016/j.est.2023.10**	**Geometry modification of a vertical**	Both papers demonstrate that geometric modifications and aspect parameters dramatically alter the

		8365	**shell-and-tube latent heat thermal energy storage system..**	melting/solidification time and the latent heat storage rate. They prove that custom domain geometries eliminate heat transfer bottlenecks, supporting your inclusion of Aspect Ratio (AR) sensitivity and geometric boundaries in your Bi-LSTM training methodology